

Context-Conditional, Cost-Asymmetric Bandwidth Allocation from Sparse, Irregularly Sampled Operator Traces

[Author name]

Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2106110@eee.buet.ac.bd

Abstract—Cellular capacity planning is usually studied as a forecasting problem and scored by RMSE. We argue this is the wrong target twice over: the decision an operator makes is an *allocation*, whose costs are asymmetric, and it must be made at a *lead time*, not one step ahead with the previous measurement in hand. Working from two ~ 900 -sample traces from Dhaka cell sites, we first show that the sampling rate of both is misread by the standard treatment: neither is hourly (86 and 99 minutes per sample), so a 24-sample “daily” lag spans 34.4 and 39.6 hours and correlates *negatively* with demand. We rebuild the evaluation protocol around the measured sampling rate, with leak-safe features, rolling-origin folds, naive baselines and Diebold–Mariano tests. We then add lead time and an allocation layer: with an asymmetric cost ratio κ , the cost-minimising allocation is the τ -quantile of the predictive distribution with $\tau^* = \kappa/(1 + \kappa)$, calibrated conformally. The advantage of a learned model over a naive forecaster *at the same lead* grows from 14% at one step to 59% at 11.5 hours. Our central contribution is that this calibration must be context-conditional: auditing the traces’ hand-labelled flags shows most do not predict the *level* of demand—including the weekend flag both source files are named after ($p = 0.77$)—but rainfall raises residual σ by 33% ($p < 0.001$). Context acts on variance, not level. A marginally calibrated allocator consequently meets its 95% target overall while delivering 86.6% [807, 912] during elevated-risk periods. At the operator’s own cost ratio, context-adaptive calibration provisions 7.5% (GP) and 14.8% (Robi) more cheaply than a fixed-margin rule tuned with hindsight. We report two negative results in full: correcting the feature design does not improve RMSE, and split-conformal coverage fails a $\pm 2\%$ check on real data (2 of 30 configurations pass) because exchangeability breaks under drift. Online conformal calibration repairs it completely, passing 30 of 30 at lead times from 1.4 to 24 hours on both operators.

Index Terms—bandwidth allocation, cellular traffic prediction, conformal prediction, irregular sampling, cost-asymmetric optimisation, service-level agreements

I. INTRODUCTION

Mobile operators provision capacity against demand they cannot observe yet. The applied literature treats this as a forecasting problem: assemble lag features, train a set of models—gradient boosting, ARIMA, an LSTM—and rank them by RMSE on a held-out split. The ranking is then reported as though the best-ranked model were the best *system*. Three things go wrong in that framing, and they compound.

The objective is not symmetric. Under-provisioning breaches a service-level agreement and degrades quality of experience; over-provisioning wastes capital. These costs differ by a factor of 5–20 in practice, and a loss function penalising them equally optimises for neither. The decision an operator actually makes is an allocation, and the cost-minimising allocation is a *quantile* of the predictive distribution rather than its mean—which changes the model class, the loss and the evaluation metric together.

The horizon is not one step. Reported evaluations almost universally predict y_t given y_{t-1} , feeding true lagged values at every step. No provisioning decision can be made that way: capacity for an interval is committed before that interval’s predecessor has been measured. This matters more than it appears, and in a direction that is not obvious. We find that at one step a learned model beats a one-line persistence baseline by 14%—a margin thin enough to invite the conclusion that the modelling is not worth its complexity—but that at an 11.5-hour lead the same model beats the naive forecaster *at that same lead* by 59%. The one-step protocol systematically understates the value of learning, because it is precisely the regime in which the trivial baseline is strongest.

The evaluation inherits assumptions from the data format. Operator traces are frequently irregularly sampled, and a design matrix indexed by sample count silently encodes a sampling rate that may be wrong. Neither trace studied here is hourly (86 and 99 minutes per sample), so the lag of 24 samples used throughout as the “daily” lag spans 34.4 and 39.6 hours and correlates *negatively* with demand. Every seasonal statement—STL decomposition, ACF peak, ARIMA(24, 1, 0)—is then made on the wrong time axis.

We address all three on two ~ 900 -sample traces from Dhaka cell sites. The small sample is a genuine constraint and we do not pretend otherwise; it is also why the methods chosen here are the right ones. Conformal calibration is distribution-free and finite-sample valid, so it needs no parametric error model that 900 points cannot support, and an explicit estimability guard makes the sample-size ceiling visible rather than letting it be silently exceeded.

Our central observation concerns the hand-labelled context-

tual annotation these traces carry—rainfall, political gatherings, power cuts, promotional offers. Such annotation for a specific Dhaka cell site cannot be downloaded, and it is the dataset’s distinctive asset; it is also, in the standard treatment, fed to models as binary dummies and never validated. Validating it shows that most flags do not predict the *level* of demand at all, including the weekend flag both source files are named after. What rainfall predicts is *dispersion*: residual σ rises 33%. Context acts on variance, not mean—and a point forecaster with a context dummy structurally cannot express that, while a context-conditional prediction interval can. This converts weak mean-predictors into useful risk signals, which is the right use of annotation that is sparse, hand-made and noisy.

Contributions.

- 1) A sampling-aware, leak-safe evaluation protocol for irregularly sampled operator traces, in which every seasonal hyperparameter is *derived* from the measured sampling rate rather than assumed (Section IV).
- 2) A multi-horizon evaluation showing that the value of learned forecasting on these traces is concentrated at operationally relevant lead times and is nearly invisible at one step (Section V).
- 3) A cost-asymmetric allocation layer with conformal calibration, evaluated on provisioning cost rather than forecast error, in which the cost ratio selects the service level at inference time rather than being trained into the model [1] (Section VI).
- 4) **Context-conditional calibration**: the observation that hand-labelled context acts on the *variance* of demand rather than its level, the demonstration that marginal calibration therefore fails inside high-risk contexts, and the repair (Section VII).
- 5) A cold-start transfer study (Section VIII) and a zero-shot foundation-model comparison showing that the sampling-rate measurement predicts, in advance, whether such a model will work at all (Section IX).

II. RELATED WORK

Cellular traffic prediction. The applied literature on mobile traffic forecasting is large and well surveyed [2]. Its dominant shape is the one we depart from: assemble lag and calendar features, train a panel of models, and rank them by RMSE or MAPE on a held-out split. Three assumptions ride along unexamined: that the loss is symmetric, that one-step-ahead error is the operational quantity, and that the sampling grid is whatever the column index implies. This paper argues all three matter, and demonstrates that the third is measurable rather than rhetorical.

Cost-aware capacity forecasting. The closest prior work to ours is DeepCog [1], [3], which makes the same central argument: an operator does not want a forecast of traffic minimising absolute error, it wants a forecast of the *capacity* to provision, and the two differ because over-provisioning and SLA violation carry different costs. DeepCog trains a three-dimensional convolutional network with a tunable asymmetric

loss and reports monetary cost rather than RMSE, reducing operating expenses by 273–381% against the best of four benchmarks at a cost ratio of $\alpha = 2$. One of those benchmarks — a fixed over-provisioning offset selected by exhaustive search over the evaluation data — is the same hindsight-tuned comparator we construct independently in Section VI-C, which we take as evidence that it is the right heuristic to measure against.

We differ in mechanism rather than in aim. DeepCog trains the asymmetry into the network’s weights, so the cost ratio is fixed at training time and a tenant with a different κ requires a different model. We leave the forecaster untouched and place the asymmetry in a calibrated quantile, so κ becomes a parameter set at inference through $\tau^* = \kappa/(1 + \kappa)$ and one model serves every cost ratio. That separation also buys a distribution-free finite-sample coverage guarantee, which a learned loss does not provide — a difference that matters little with DeepCog’s metropolitan-scale measurements over 470 base stations and a great deal at $n \approx 900$. Their evaluation is at a five-minute horizon on aggregated demand; ours is at lead times to 24 hours on single sites.

SLA-aware prediction and slice resource allocation. Also close to our framing is the work of Tuna and Soysal [4], [5], which predicts NextG slice traffic under explicit SLA violation constraints rather than under squared error, and observes—as we do—that a symmetric loss is the wrong objective when breaching a service level and wasting capacity cost different amounts. MicroOpt [6] goes further into the decision layer, optimising slice resources against a learned model of SLA satisfaction. We differ in three ways. Those systems train an asymmetric or constrained *point* predictor; we keep an ordinary point model and move the asymmetry into a calibrated quantile, which decouples the forecaster from the cost ratio and lets κ change without retraining. Their guarantees are empirical over large regularly sampled datasets; ours are distribution-free and finite-sample, which is what ~ 900 irregularly sampled points can actually support. And neither conditions the service level on operating context, which is our central contribution.

Conformal prediction. We build directly on the conformal framework [7], [8]: given exchangeable residuals, a calibration quantile yields finite-sample marginal coverage with no distributional assumption. Conformalized quantile regression [9] makes interval width respond to local difficulty, and Mondrian conformal prediction partitions the calibration set into groups to obtain per-group validity. Exact conditional coverage is impossible without assumptions, and Gibbs, Cherian and Candès [10] characterise what is achievable against a specified class of shifts. Section VII is an applied instance of that tension: the groups are not statistical constructs but hand-labelled operating conditions—rainfall, power cuts—and the finding is that a marginally valid allocator is systematically invalid inside the group where the network is most stressed. Separately, exchangeability itself fails on a drifting two-month trace; adaptive conformal inference [11] updates the requested level online from realised breaches and restores long-run

coverage without assuming exchangeability at all.

Time-series foundation models. Chronos [12] tokenises scaled time-series values and trains a language-model architecture over them; TimesFM [13] is a decoder-only model pretrained on a large corpus. Both are marketed on zero-shot transfer to unseen series. Neither consumes timestamps—they read a bare sequence of values—which is inconsequential on a regular grid and, we show in Section IX, decisive on an irregular one. We are not aware of prior work linking zero-shot foundation-model performance to the misregistration between a trace’s sampling interval and its dominant period, and we offer that as a cheap pre-deployment diagnostic.

Evaluation methodology. Our protocol follows standard forecasting practice that the applied cellular literature frequently omits: rolling-origin evaluation rather than a single split [14], scale-free error reporting against a naive baseline [15], and Diebold–Mariano tests [16] with Benjamini–Hochberg false-discovery control [17] when ranking a panel of models. None of this is novel. It is included because applying it to these traces reverses several published conclusions.

III. DATA

TABLE I
THE TWO OPERATOR TRACES.

	Grameenphone	Robi Dhaka-3
Observations	905	888
Span (days)	55	62
Median sampling interval	86 min	99 min
Samples per day	~17	~15
Mean demand	92.9 Gbps	145.5 Gbps
Contextual flags	9	5

Both traces carry hand-annotated local context (`is_rain`, `is_political_gathering`, `is_offer`, `is_powercut`, `is_drama`, and others). This annotation is the distinctive asset of the dataset—it is not something that can be downloaded for a Dhaka cell site—and Section VII is built on it.

IV. A CORRECTED EVALUATION PROTOCOL

A. The traces are not hourly

Both traces are treated as hourly in the standard treatment, so a lag of 24 samples is used as the daily lag. At 86 min per sample that lag spans 34.4 hours and sits roughly anti-phase to the daily cycle.

The magnitude of this is not particular to our traces. On the Telecom Italia Milan measurements, Mehri *et al.* [18] report $\text{ARIMA}(3,0,4)$ at an MSE of 513.53 against $\text{SARIMA}(1,0,1)(1,0,1,144)$ at 61.78 — an $8.3\times$ improvement obtained by setting the seasonal period to 144 steps, which is exactly 24 hours at their 10-minute sampling interval. The same mechanism, on a different operator’s data, at a different sampling rate.

Every seasonal statement made on the sample-count axis—STL decompositions, ACF and FFT

TABLE II
AUTOCORRELATION AT THE MEASURED DAILY PERIOD VERSUS AT LAG 24, AND THE SEASONAL-NAIVE RMSE PENALTY FOR USING THE LATTER.

	Measured period	At lag 24
<i>Autocorrelation</i>		
GP (period 17)	+0.451	−0.347
Robi (period 15)	+0.822	−0.383
<i>Seasonal-naive RMSE (penalty)</i>		
GP	18.16	28.42 (+56%)
Robi	26.58	74.12 (+179%)

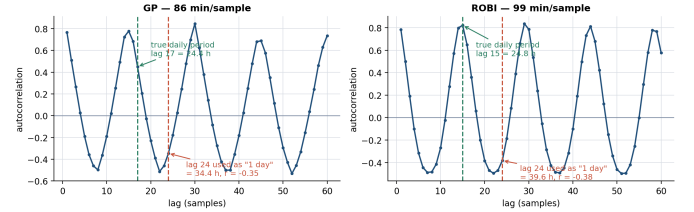


Fig. 1. Autocorrelation by lag, with the measured daily period and lag 24 marked. The feature used throughout the standard treatment as the daily lag correlates negatively with demand on both traces.

peaks, $\text{AutoReg}(\text{lags}=24)$, $\text{ARIMA}(24,1,0)$, $\text{ExponentialSmoothing}(\text{seasonal_periods}=24)$ —is stated on the wrong axis. In this work the daily period is measured per operator and every seasonal hyperparameter derives from it, while seasonality itself is carried by Fourier terms in wall-clock time, which are exact under irregular sampling.

B. Target leakage

Rolling statistics computed without a preceding shift place y_t inside its own feature vector. Restoring the shift moves the best GP result from 6.54 to 19.30 RMSE (+195%) and Robi from 16.32 to 27.37 (+68%). The corrected GP figure is worse than predicting the training mean (18.87). We enforce this structurally: `assert_no_leakage` perturbs the final target value and asserts that no feature column moves, and the benchmark refuses to run if it fails.

C. Baselines, folds, and significance

All models share one rolling-origin fold schedule (8 expanding-window folds), every table reports naive baselines (persistence, seasonal-naive at the measured period, drift, train-mean) and MASE alongside RMSE, and rankings carry Diebold–Mariano p -values under Benjamini–Hochberg correction.

D. A negative result: the correction does not improve accuracy

The feature ablation is reported in full because it cuts against the correction. On both operators, a faithful reproduction of the original feature design (with the leak removed) is within noise of, and on Robi slightly better than, the corrected design.

Tree ensembles route around a mis-specified `lag_24` by leaning on the short lags, so the sampling-rate error costs almost nothing in RMSE. Its cost is **interpretive**: every

TABLE III
CORRECTED ONE-STEP BENCHMARK (RMSE, MEAN $\pm$ SD ACROSS 8 FOLDS).

Model	GP	Robi
Random forest	10.41 $\pm$ 1.19	20.75 $\pm$ 4.42
XGBoost	10.80	21.32
Ridge	11.26	22.72
Persistence	12.13	29.75
Seasonal-naive (measured)	18.31	26.26
Seasonal-naive at lag 24	29.74	73.51

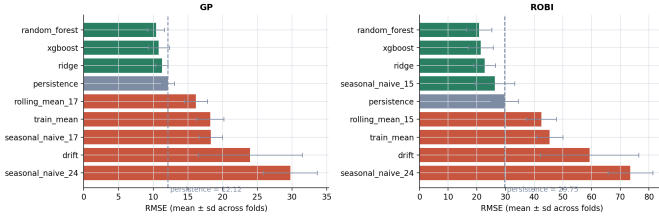


Fig. 2. Corrected benchmark with fold spreads and the persistence line.

seasonal claim in the standard treatment is stated on the wrong axis, and Section IV-A shows what the error costs a model that cannot route around it. We report this rather than bury it.

V. LEAD TIME

Every result above, and every result in the studies this extends, predicts y_t with y_{t-1} in hand. No network provisions from that: the capacity decision for an interval is made before the previous measurement arrives.

We evaluate direct multi-horizon models at 1.5–24 h. Three details make the evaluation honest.

Feature roles. History columns (lags, rolling statistics) are read at the forecast *origin*; Fourier and calendar terms, being deterministic functions of wall-clock time, are read at the *target*. An operator provisioning for 9 p.m. knows it will be 9 p.m.

Per-horizon baselines. An h -step model is scored against an h -step naive forecaster. Comparing a 24-hour-ahead model to one-step persistence is how multi-step results are most often overstated.

Embargo. Direct training pairs carry targets $h - 1$ rows after their origin, so those rows are dropped from fitting and calibration; otherwise the model is trained on outcomes that had not occurred when the first test forecast was issued.

TABLE IV
FEATURE ABLATION (RMSE). THE CORRECTED DESIGN DOES NOT WIN.

Configuration	GP	Robi
Lags only	9.98	21.78
Original design (leak removed)	10.11	20.53
Original design, lag corrected	10.12	21.39
Full corrected design	10.41	20.75

TABLE V
GP: RMSE AGAINST FORECAST LEAD TIME, 8 FOLDS.

Lead	1.43 h	2.87 h	5.73 h	11.47 h	24.37 h
Random forest	10.40	11.85	12.48	12.83	12.82
Naive at lead	12.12	17.41	25.11	31.16	18.26
Advantage	−14%	−32%	−50%	−59%	−30%

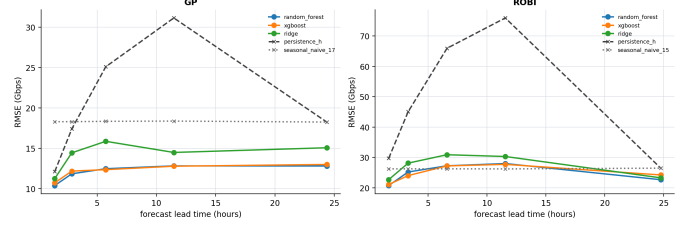


Fig. 3. Accuracy against lead time, both operators. The learned model degrades far more slowly than the naive forecaster at the same lead.

Robi replicates it: -30% at 1.65 h growing to -63% at 11.55 h. Across a $17\times$ longer horizon the learned model degrades 23% while the naive baseline degrades 157%.

This is the central empirical claim of the paper. The value of learning on these traces is not visible at one step; it is visible at the lead time an allocator needs. The conventional one-step protocol measures these models in the single regime where they look worst.

Two secondary findings follow. *Direct beats recursive on GP* and the gap widens with horizon (13.01 versus 24.18 at a 24-hour lead, $+86\%$); on Robi the two are within noise at intermediate horizons, so the honest statement is that direct is the safer default rather than a universal winner. And *on Robi, forecasting 24 hours ahead with yesterday's value at the same hour (26.49) beats forecasting 1.6 hours ahead with the latest observation (29.40)*. The daily cycle there is stronger than short-run persistence. This inverts the usual decay-with-horizon intuition and is visible only once the daily period is measured as 15 samples rather than assumed to be 24.

VI. ALLOCATION

A. Cost model

For an allocation A against realised demand y , with over-provisioning unit cost c_o and under-provisioning κ times more expensive,

$$C = c_o (A - y)^+ + \kappa c_o (y - A)^+. \quad (1)$$

The cost-minimising allocation is the τ -quantile of the predictive distribution with

$$\tau^* = \frac{\kappa}{1 + \kappa}, \quad (2)$$

so the operator's cost ratio *selects the quantile*: $\kappa = 10$ gives $\tau^* = 0.909$. We verify this correspondence empirically rather than only asserting it.

B. Calibration

Quantile forecasts come from gradient-boosted quantile regression under the pinball loss [19]. They are calibrated by split conformal prediction, which is distribution-free and finite-sample valid—the right choice at $n \approx 900$, where no parametric error model is credible. An estimability guard refuses any τ a calibration set cannot express (a set of m residuals cannot express a level finer than $1/(m+1)$; $\tau = 0.99$ needs 99) rather than silently returning the largest residual.

C. What to measure, and a comparison that does not work

The natural comparison—capacity required at an equal SLA violation rate, against the $A = 1.3\hat{y}$ rule operators use—**does not favour our method and we do not claim it**. No conformal family reaches a 1%, 2% or 5% violation target on either trace. The reason is that the comparison is not well posed: drawing the fixed-margin frontier requires knowing which margin achieved which violation rate *on the test data*, and “the margin that lands at 1% violations” is a choice no operator can make in advance. Conformal picks its level a priori from κ . Scoring an a-priori method against a hindsight-tuned heuristic on the heuristic’s home metric is not a fair test. We also implemented a multiplicative conformal variant, on the hypothesis that an additive-versus-multiplicative mismatch explained the gap; it did not (−3.1% against −2.7%), which is what rules that explanation out. The hindsight-tuned fixed margin we compare against below is not a strawman of our own construction: it is the MAE-post-best baseline of [1], arrived at there by the same reasoning.

TABLE VI
PROVISIONING COST AT $\kappa = 10$, AGAINST A FIXED-MARGIN RULE GIVEN ITS BEST HINDSIGHT-TUNED MARGIN—A SETUP BIASED AGAINST US.

Method ($\kappa = 10$)	Cost at τ^*	vs tuned
GP, context-adaptive	19.27	−7.5%
GP, Mondrian	20.11	−3.5%
GP, marginal	21.35	+2.4% (<i>worse</i>)
GP, fixed margin (tuned)	20.84	—
GP, static peak	29.52	+41.6% (<i>worse</i>)
Robi, context-adaptive	49.34	−14.8%
Robi, marginal	59.09	+2.0% (<i>worse</i>)
Robi, fixed margin (tuned)	57.93	—

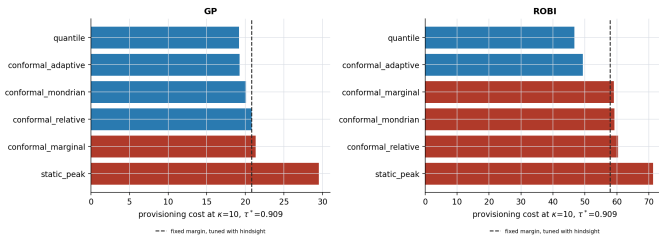


Fig. 4. Provisioning cost by calibration family, both operators.

Read the pattern, not the winner: **conditioning the margin on predicted uncertainty beats the tuned heuristic on both**

operators; calibrating one margin for all conditions loses to it on both. That is Section VII’s idea, measured in money.

VII. CONTEXT-CONDITIONAL CALIBRATION

A. Context acts on variance, not level

Welch t -tests on the level and Levene tests on the spread of hour-detrended residuals, GP:

TABLE VII
CONTEXT-FLAG AUDIT ON GP. SIX OF NINE FLAGS ARE INDISTINGUISHABLE FROM NOISE ON THE MEAN; ONE RAISES THE RESIDUAL SPREAD BY A THIRD.

Flag	n	Δ mean	p (level)	σ ratio	p (var.)
is_weekend	261	+0.37	0.77	0.96	ns
is_event	275	−0.23	0.86	1.02	ns
is_drama	147	−1.52	0.34	1.00	ns
is_powercut	118	−2.63	0.13	1.02	ns
is_political	101	+2.69	0.17	1.07	ns
is_rain	131	+4.50	0.026	1.330	< 0.001
is_offer	235	−5.73	5.1e-06	0.90	ns

Two things follow. Six of the nine flags are indistinguishable from noise on the mean, including the weekend flag that **both source filenames advertise**. And **is_rain** raises residual σ from 16.4 to 21.8 Gbps—a 33% increase in *uncertainty*—while moving the mean only modestly.

A point forecaster with a context dummy cannot express that. A context-conditional prediction interval can. This converts weak mean-predictors into useful **risk** signals: the flags need not predict demand well, only mark when the forecast is less trustworthy.

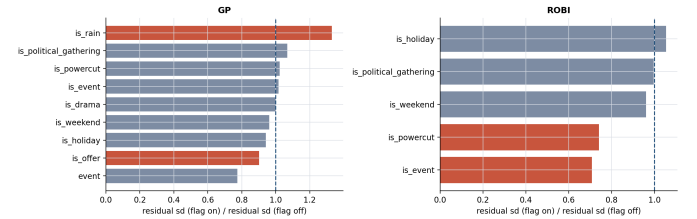


Fig. 5. Context-flag audit: effect on level against effect on residual spread.

B. The failure, and the repair

Groups are formed by priority to keep them disjoint, and merged into two—*elevated risk* = $\text{rain} \cup \text{gathering}$ (199 rows) against *baseline* (689)—because a four-group split leaves the gathering group with ~ 20 calibration residuals. The baseline group carries $n = 354$ test points and the elevated-risk group $n = 179$; exact Clopper–Pearson intervals are used rather than a normal approximation, which at $n = 179$ is not reliable.

The failure is established; the repair is directional. These two halves of the claim have different evidential status and we separate them.

The failure is decisive. At $\tau = 0.90$ and $\tau = 0.95$ the marginal interval inside the elevated-risk group excludes the nominal level ([.702, .830] against 0.90; [.807, .912] against 0.95). A marginally calibrated allocator demonstrably does not

TABLE VIII

GP: COVERAGE POOLED ACROSS FOLDS, WITH CLOPPER–PEARSON 95% INTERVALS. THE FAILURE OF MARGINAL CALIBRATION INSIDE THE ELEVATED-RISK GROUP IS DECISIVE; THE CONTEXT-CONDITIONAL REPAIR IS DIRECTIONAL.

Nominal τ	Group	Marginal	Locally adaptive	Mondrian	Online (ACI)
0.80	baseline	0.831 [.787, .868]	0.814 [.769, .853]	0.799 [.754, .840]	0.825 [.781, .863]
0.80	elevated	0.693 [.620, .759]	0.726 [.655, .790]	0.732 [.661, .795]	0.760 [.690, .820]
0.90	baseline	0.915 [.881, .942]	0.887 [.849, .918]	0.887 [.849, .918]	0.901 [.865, .930]
0.90	elevated	0.771 [.702, .830]	0.816 [.751, .870]	0.788 [.720, .845]	0.911 [.859, .948]
0.95	baseline	0.958 [.931, .976]	0.929 [.898, .954]	0.944 [.914, .965]	0.949 [.921, .970]
0.95	elevated	0.866 [.807, .912]	0.899 [.846, .939]	0.933 [.886, .965]	0.944 [.900, .973]

deliver its promised service level when the network is under stress.

The context-conditional repair is not decisive at this sample size. Locally adaptive calibration moves the $\tau = 0.95$ elevated-risk estimate from 0.866 to 0.899, but the intervals overlap substantially and the adaptive interval still excludes 0.95. Mondrian does better (0.933, [.886, .965], which does contain the nominal level) but on the same 179 points. The point estimates favour the context-conditional methods at every level on both groups; the intervals do not separate them from marginal calibration. This is a limit of the data, not of the method: 179 elevated-risk test points cannot resolve a 3–7 point coverage difference.

C. The prediction is falsifiable, and it is null on Robi

The audit finds **no Robi flag with elevated residual variance**—`is_powercut` and `is_event` mark *calmer* periods—and Robi carries no rain annotation. The method should therefore do nothing there, and it does: at $\tau = 0.95$ the elevated-risk group sees 0.889 (marginal), 0.875 (adaptive), 0.903 (Mondrian).

We state the claim in this form: *context-conditional calibration helps exactly when context carries variance signal, and whether it does can be tested in advance.* A falsifiable prediction confirmed in both directions on two operators is stronger than “it works”.

D. A caveat on the cost result

On Robi the adaptive cost gain of Section VI-C cannot be credited to context. The uncertainty model learns $\hat{\sigma}(x)$ from the whole feature vector, so where no flag carries variance signal the adaptivity comes from time and lag features. The supportable claim is that *conditioning the margin on predicted uncertainty pays*; only on GP is that uncertainty demonstrably contextual.

VIII. COLD START

How much of its own history does a newly deployed site need? We run four arms on a fixed GP test window with Robi as the source. This is possible only because Section IV defines lags in wall-clock hours: at 86 and 99 min per sample, a design matrix indexed by sample count is not comparable across sites.

TABLE IX

COLD START AT A 6-HOUR LEAD (PERSISTENCE = 26.49). RMSE BY DAYS OF THE NEW SITE’S OWN HISTORY.

Days of local history	3	5	7	14	21
Own data alone	20.61	19.81	18.98	14.82	15.86
Pretrained + fine-tuned	19.65	19.87	19.91	15.63	17.45
Pretrained, no fine-tuning	20.39	20.01	19.74	19.06	19.04

A model trained entirely on the other operator, given only enough history to fix the new site’s level and scale, is 23% better than persistence with **zero** training data of its own. Pretraining then helps only while the site is data-poor: +4.6% at 3 days, nothing by 5–7, and –10% by 21 days where the borrowed weights hold the model back. **The crossover is**

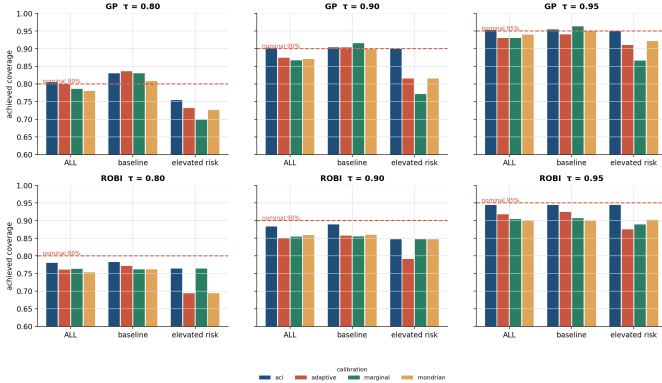


Fig. 6. Per-context coverage, both operators, three levels.

An unexpected result. The largest and only clearly decisive improvement inside the elevated-risk group comes from *online* calibration, which is not context-conditional at all: at $\tau = 0.90$ it reaches 0.911 [.859, .948] against marginal’s 0.771 [.702, .830]—non-overlapping intervals. Adaptive conformal inference adapts to *whatever* is currently causing breaches, and elevated-risk periods are clustered in time, so an online level update partly absorbs them without ever being told the context. The honest reading is that the two mechanisms are complementary and that on this dataset the temporal one is better evidenced than the contextual one. A per-group online update—the obvious combination—is untried here and is the clearest next step. The capacity cost of all of this is near-identical ($\sim 1.18\times$ demand at $\tau = 0.95$).

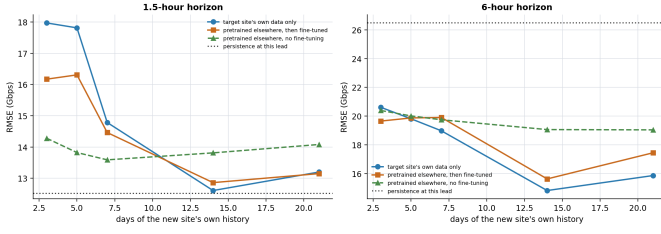


Fig. 7. Cold-start curves at both horizons.

about a week. At a 1.5-hour lead the study answers nothing—no arm beats persistence—which is a second demonstration of Section V.

IX. ZERO-SHOT FOUNDATION MODELS

With ~ 900 samples per site, is bespoke per-operator training worth it, or does a model that has never seen this network do just as well? We evaluate Chronos-Bolt Small strictly zero-shot—no fitting, no fine-tuning, not even a scaling constant—on the same fold schedule, lead times and test blocks as Section V. It runs on CPU in about seven minutes.

A. The answer depends entirely on the operator

TABLE X
ZERO-SHOT CHRONOS-BOLT AGAINST THE BEST TRAINED MODEL AND AGAINST PERSISTENCE.

Lead	Zero-shot	Best trained	vs trained	vs pers.
<i>GP</i>				
1.4 h	11.18	10.40	+7.5%	−7.8%
2.9 h	13.53	11.85	+14.2%	−22.3%
5.7 h	13.50	12.36	+9.2%	−46.2%
11.5 h	14.41	12.79	+12.7%	−53.8%
24.4 h	15.20	12.82	+18.6%	−16.8%
<i>Robi</i>				
1.7 h	30.54	20.77	+47.1%	+2.7%
3.3 h	38.84	24.04	+61.5%	−13.8%
6.6 h	40.34	27.23	+48.2%	−38.8%
11.6 h	42.56	27.74	+53.4%	−44.0%
24.8 h	43.63	22.69	+92.3%	+64.7%

On GP the zero-shot model is a credible system: it beats persistence at every lead, by as much as 54%, and trails a model trained on the site’s own history by only 7.5–18.6%. On Robi it trails by 47–92% and is *worse than persistence* at both the shortest and the longest lead.

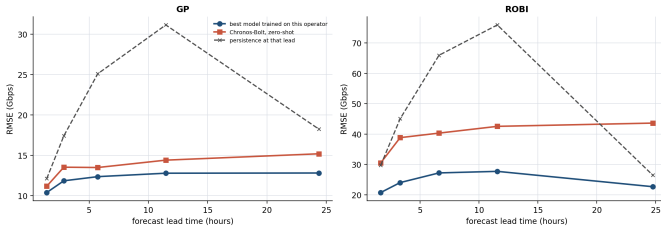


Fig. 8. Zero-shot against trained and naive forecasters, both operators.

B. Why, and it is predictable in advance

The gap is not random, and Section IV explains it. A foundation model consumes a bare sequence of values with no timestamps. It cannot be told that a step is 86 minutes on one trace and 99 on the other, so it must infer periodicity from the sequence itself—and on an irregularly sampled trace the daily cycle does not occupy a whole number of steps.

TABLE XI
MISREGISTRATION BETWEEN THE SAMPLING GRID AND THE DAILY CYCLE.

	Gap	Exact daily period	Drift/day
GP	86 min	16.744 samples	0.256
Robi	99 min	14.545 samples	0.455

Robi’s cycle slips almost twice as fast. Over the 55-day span that is ~ 25 samples of accumulated drift, about 1.7 complete cycles, against GP’s ~ 14 . A sequence model with no clock cannot hold phase against that, and the consequence is exactly where theory says it should be: **the worst result of the entire experiment is Robi at the 24-hour horizon** (+92% against trained, +65% against persistence), the one setting where the daily cycle is the whole signal. The naive baseline wins there simply by stepping back one *measured* cycle—the thing the foundation model cannot do.

This makes the sampling-rate correction more than a repair of prior work. It is a *diagnostic*: measuring the sampling interval tells an operator in advance whether a timestamp-blind foundation model will work on their trace. We are not aware of this being stated in the zero-shot forecasting literature, and it is cheap to check.

C. Its quantiles cannot be used for provisioning as they come

Chronos-Bolt is quantile-native, so it reaches the allocation layer with no quantile-regression step. But its quantile head was trained on levels 0.1 to 0.9 only, and a request above that is silently clipped—a $\tau = 0.95$ request returns the $\tau = 0.90$ numbers unchanged. Since $\tau^* = \kappa / (1 + \kappa)$, a ceiling of 0.9 corresponds to a cost asymmetry of only $\kappa = 9$. An operator for whom under-provisioning costs $20\times$ more needs $\tau^* = 0.952$ and cannot obtain it from the model at all.

TABLE XII
ZERO-SHOT QUANTILE CALIBRATION, AND ITS REPAIR BY SPLIT CONFORMAL.

Op.	τ	Zero-shot	Conformalised	Capacity
GP	0.80	0.791	0.815	1.112 $\rightarrow$ 1.124
GP	0.90	0.886	0.903	1.166 $\rightarrow$ 1.183
GP	0.95	0.886 (<i>clip</i>)	0.941	1.166 $\rightarrow$ 1.225
Robi	0.80	0.662	0.813	1.148 $\rightarrow$ 1.245
Robi	0.90	0.786	0.909	1.230 $\rightarrow$ 1.328
Robi	0.95	0.786 (<i>clip</i>)	0.950	1.230 $\rightarrow$ 1.389

On Robi the zero-shot 80% interval delivers 66%. Split-conformal calibration on held-out residuals repairs every level on both traces, including the ones the model cannot express

natively. So the allocation machinery is not an alternative to the foundation model; it is what makes the foundation model deployable. That is the useful synthesis: **a zero-shot forecaster supplies the point prediction cheaply, and conformal calibration supplies the service-level guarantee it cannot provide itself.**

X. A NEGATIVE RESULT ON COVERAGE, AND ITS REPAIR

Enforcing the $\pm 2\%$ coverage check on real backtest output rather than synthetic data: **249 of 360 configurations fail, 232 of them by under-covering; of the 30 marginal configurations, only 2 pass.** The one-sidedness is diagnostic. Split conformal’s guarantee is conditional on exchangeability, and a 55-day trace with a trend does not supply it.

We therefore apply adaptive conformal inference, updating the requested miscoverage online from realised breaches,

$$\alpha \leftarrow \alpha + \gamma (\alpha_{\text{target}} - \text{err}), \quad (3)$$

whose long-run coverage converges without any exchangeability assumption. All feedback is strictly causal: the outcome at t affects only allocations from $t + 1$.

TABLE XIII
MARGINAL COVERAGE ACROSS 5 LEAD TIMES, 3 LEVELS AND BOTH OPERATORS—30 CONFIGURATIONS.

Method	Passing $\pm 2\%$	GP gap	Robi gap
Online (ACI)	30 / 30	+0.2 pp	−0.9 pp
Locally adaptive	6 / 30	−2.8 pp	−4.2 pp
Mondrian	5 / 30	−3.3 pp	−5.5 pp
Marginal	2 / 30	−3.4 pp	−4.9 pp

TABLE XIV
GP: ACHIEVED COVERAGE BY LEAD TIME. ONLINE CALIBRATION HOLDS ITS TARGET AS THE FORECAST DEGRADES; STATIC CALIBRATION DOES NOT.

τ	1.4 h	2.9 h	5.7 h	11.5 h	24.4 h
0.90, online	0.904	0.902	0.906	0.907	0.901
0.90, marginal	0.867	0.878	0.851	0.839	0.868
0.95, online	0.947	0.946	0.949	0.945	0.950
0.95, marginal	0.927	0.957	0.915	0.909	0.924

Online calibration holds its nominal service level in all 30 (operator, lead time, level) configurations, while static calibration meets it in 2 to 6 and drifts as far as 7 points below nominal as the forecast degrades. For a deployable allocator this is the property that matters.

This is *marginal* coverage, and we do not overstate it: counting per-group rows as well, ACI passes 61 of 90 configurations against 12–20 for the static methods. ACI is not group-conditional, so it does not subsume Section VII—the two repairs address different failures and are complementary. A per-group online update is the obvious next step and is untried here.

XI. LIMITATIONS

905 and 888 observations from two cell sites over ~ 2 months. The sample size does not support strong claims about model superiority, and the span cannot speak to seasonal or annual effects. Context flags are hand-labelled by a single annotator with no inter-rater check. Robi carries five of the nine flags and no rain annotation, so its context-conditional results rest on an 89-row group and are directional only.

Two limits are measured rather than asserted, and both are reported above: correcting the feature design does not improve accuracy (Section IV-D), and split-conformal coverage fails its $\pm 2\%$ target on these traces (Section X). Absolute coverage guarantees should not be claimed from the static conformal results; the context-conditional comparison is unaffected, being a relative comparison between methods calibrated on identical data.

The methods here are chosen to be robust to those limits rather than to hide them: conformal calibration is distribution-free and finite-sample valid, the estimability guard makes the sample-size ceiling explicit, and the drift that breaks exchangeability is handled by a method that does not assume it.

XII. CONCLUSION

Treating capacity planning as a forecasting problem scored by RMSE gets three things wrong at once, and correcting each changes what the data says.

Correcting the **horizon** rescues the case for learning. At one step a learned model beats persistence by 14% on GP, a margin that would not justify deploying anything; at the lead time a provisioning decision actually requires, it beats the naive forecaster at that same lead by 59%, and by 63% on the second operator. The conventional protocol measures these systems in the one regime where they look worst.

Correcting the **objective** makes the comparison meaningful. Scored on provisioning cost at the level the theory prescribes from the operator’s own cost ratio—against a fixed-margin heuristic given its best margin with hindsight—context-adaptive conformal allocation is 7.5% (GP) and 14.8% (Robi) cheaper, while marginally calibrated allocation is *worse* than that heuristic on both traces. The saving comes specifically from conditioning the margin on predicted uncertainty.

Correcting the **sampling assumption** is what makes anything comparable across sites, and it pays two unanticipated dividends. Because lags are defined in wall-clock hours rather than sample counts, a model trained on one operator transfers to the other: given only enough history to fix a new site’s scale, a borrowed model is 23% better than persistence with zero training data of its own, and the crossover to training locally arrives after about a week. And the same measurement turns out to *predict* when a zero-shot foundation model will fail—such a model reads a bare sequence with no clock, so it cannot hold phase against a daily cycle that occupies 14.545 samples, and it duly collapses on the trace where that misregistration is worst while remaining competitive on the trace where it is not.

On the context-conditional claim we are deliberately careful. That marginal calibration *fails* inside high-variance contexts is established: at $\tau = 0.95$ its elevated-risk interval is $[.807, .912]$, excluding the nominal level. That context-conditional calibration *repairs* it is directional—the point estimates improve at every level, but 179 elevated-risk test points cannot resolve a 3–7 point difference. The largest decisive improvement on that group in fact comes from online calibration, which is not context-conditional at all, because elevated-risk periods cluster in time. A per-group online update is the obvious combination and is untried.

Two negative results are reported in full because they discipline the claims: the corrected feature design does not improve RMSE, since tree ensembles route around a misspecified lag; and static conformal coverage fails its $\pm 2\%$ target almost everywhere, because a 55-day trace with a trend does not satisfy exchangeability. Online calibration repairs the latter completely—30 of 30 configurations—and that is the property a deployable allocator needs.

None of this depends on the dataset being large, which is what makes it defensible at $n = 900$. What the sample size does foreclose is any strong claim about model superiority, and we make none.

REPRODUCIBILITY

Fixed seeds throughout. 47 verification gates encode the specific failures corrected here—leakage, the sampling-rate assumption, the inverted split, the estimability bound, the multi-horizon embargo, the causality of the online update—so a refactor cannot silently reintroduce them. Every figure is regenerated from the stored result tables alone, so the paper rebuilds without refitting any model.

REFERENCES

- [1] D. Bega, M. Gramaglia, M. Fiore, A. Banchs, and X. Costa-Perez, “DeepCog: Optimizing resource provisioning in network slicing with AI-based capacity forecasting,” *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 2, pp. 361–376, 2020, extended version of the INFOCOM 2019 paper.
- [2] X. Wang *et al.*, “A survey on deep learning for cellular traffic prediction,” *Intelligent Computing*, vol. 3, p. 0054, 2024, full author list to be completed from the published PDF.
- [3] D. Bega, M. Gramaglia, M. Fiore, A. Banchs, and X. Costa-Perez, “DeepCog: Cognitive network management in sliced 5G networks with deep learning,” in *IEEE INFOCOM 2019 – IEEE Conference on Computer Communications*, 2019, pp. 280–288.
- [4] E. Tuna and A. Soysal, “Multivariate and multi-step traffic prediction for NextG networks with SLA violation constraints,” *arXiv preprint arXiv:2304.11156*, Apr. 2023.
- [5] —, “Multivariate, multi-step, and spatiotemporal traffic prediction for NextG network slicing under SLA constraints,” *arXiv preprint arXiv:2309.03898*, Sep. 2023.
- [6] M. Sulaiman, M. Ahmadi, B. Sun, N. Saha, M. A. Salahuddin, R. Boutaba, and A. Saleh, “MicroOpt: Model-driven slice resource optimization in 5G and beyond networks,” *arXiv preprint arXiv:2407.18342*, Jul. 2024.
- [7] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.
- [8] A. N. Angelopoulos, R. F. Barber, and S. Bates, “Theoretical foundations of conformal prediction,” *arXiv preprint arXiv:2411.11824*, Nov. 2024, to be published by Cambridge University Press.
- [9] Y. Romano, E. Patterson, and E. J. Candès, “Conformalized quantile regression,” in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [10] I. Gibbs, J. J. Cherian, and E. J. Candès, “Conformal prediction with conditional guarantees,” *arXiv preprint arXiv:2305.12616*, May 2023.
- [11] I. Gibbs and E. J. Candès, “Adaptive conformal inference under distribution shift,” in *Advances in Neural Information Processing Systems 34 (NeurIPS)*, 2021.
- [12] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiesner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, “Chronos: Learning the language of time series,” *arXiv preprint arXiv:2403.07815*, Mar. 2024.
- [13] A. Das, W. Kong, R. Sen, and Y. Zhou, “A decoder-only foundation model for time-series forecasting,” *arXiv preprint arXiv:2310.10688*, Oct. 2023.
- [14] L. J. Tashman, “Out-of-sample tests of forecasting accuracy: an analysis and review,” *International Journal of Forecasting*, vol. 16, no. 4, pp. 437–450, 2000.
- [15] R. J. Hyndman and A. B. Koehler, “Another look at measures of forecast accuracy,” *International Journal of Forecasting*, vol. 22, no. 4, pp. 679–688, 2006.
- [16] F. X. Diebold and R. S. Mariano, “Comparing predictive accuracy,” *Journal of Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.
- [17] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: A practical and powerful approach to multiple testing,” *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.
- [18] H. Mehri, H. Chen, and H. Mehrpouyan, “Cellular traffic prediction using online prediction algorithms,” *arXiv preprint arXiv:2405.05239*, May 2024.
- [19] R. Koenker and G. Bassett, “Regression quantiles,” *Econometrica*, vol. 46, no. 1, pp. 33–50, 1978.